

Research Statement

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I am an economist interested in designing and studying resource allocation algorithms that perform well in both theory and practice. While I am a theorist by training, my interests are broader: I have three experimental papers and five applied microeconomics papers using data. In general, I like to think of problems at a high-level and then apply the techniques that I think are most useful for a concrete problem, rather than forcing a specific approach.

My work falls into three connected topics:

1. *The cost of perfect desiderata.* When designing a resource allocation mechanism, a number of desirable goals are unfortunately incompatible. My work quantifies how much we lose in terms of efficiency when we restrict ourselves to some of these, such as strategy-proofness or stability (Ortega and Klein, 2023; Ortega et al., 2026b). Given the large efficiency losses caused by stability we documented, my work suggests how to improve the efficiency of the celebrated Deferred Acceptance mechanism through justifiable priority violations (Ortega and Arribillaga, 2026), and what such stable-dominating mechanisms can and cannot achieve (Ortega et al., 2026c,a; Knipe and Ortega, 2026). And given the large efficiency losses caused by strategy-proofness, my work explores how to achieve weaker non-manipulability properties that prevent strategic play by cognitively limited agents (Ortega and Segal-Halevi, 2022).
2. *Merging markets.* Matching markets are often fragmented, so my research asks who, if any, benefits and loses if they were to integrate? The answer is not obvious, as merging markets leads to more choice but also more competition. Yet, the majority of agents benefit from integration, albeit the overall losses can be substantial (Ortega, 2018), as I showed in my first paper (called “*seminal*” by Kamada and Kojima (2026)). A natural application of my work is school district consolidation. Using Budapest data, we show that integrated school choice yields large welfare gains, equivalent to students attending a school five kilometres closer (Klein et al., 2024).
3. *Great in theory, great in practice?* Economic theorists and computer scientists have come up with beautiful algorithms to divide heterogeneous divisible goods like land or to assign goods without money, like school seats, that have nice fairness guarantees. But do these beautiful, yet complex algorithms work well in practice? My work has shown that complex envy-free mechanisms actually work quite well in the lab, despite numerous manipulations and mistakes (Kyropoulou et al., 2022). It also has shown that when we test these algorithms as Deferred Acceptance on their actual users, like parents, rather than undergraduate students, we learn new aspects of how real users interact with the algorithm and the nature of the mistakes they make, guiding insights on the external validity of the literature (Freer et al., 2026). Finally, it has shown that missing details, such as the possibility of appealing the allocation (as done in England annually by over fifty thousand students), changes how students report their preferences and the equilibrium outcomes generated by well-known mechanisms (Cerrone et al., 2026).

In what follows, I explain each of these three research areas in detail, as well as provide a brief overview of my most exciting current projects.

The Cost of Perfect Desiderata

Two of my favourite papers quantify the expected average rank loss in school choice when we restrict to strategy-proof mechanisms (Ortega and Klein, 2023, *Games and Economic Behavior*) or to stable ones (Ortega et al., 2026, *Theoretical Economics*). The way to do this, at the start, is not to consider the performance of a mechanism in a specific cherry-picked instance, but its expected behavior when

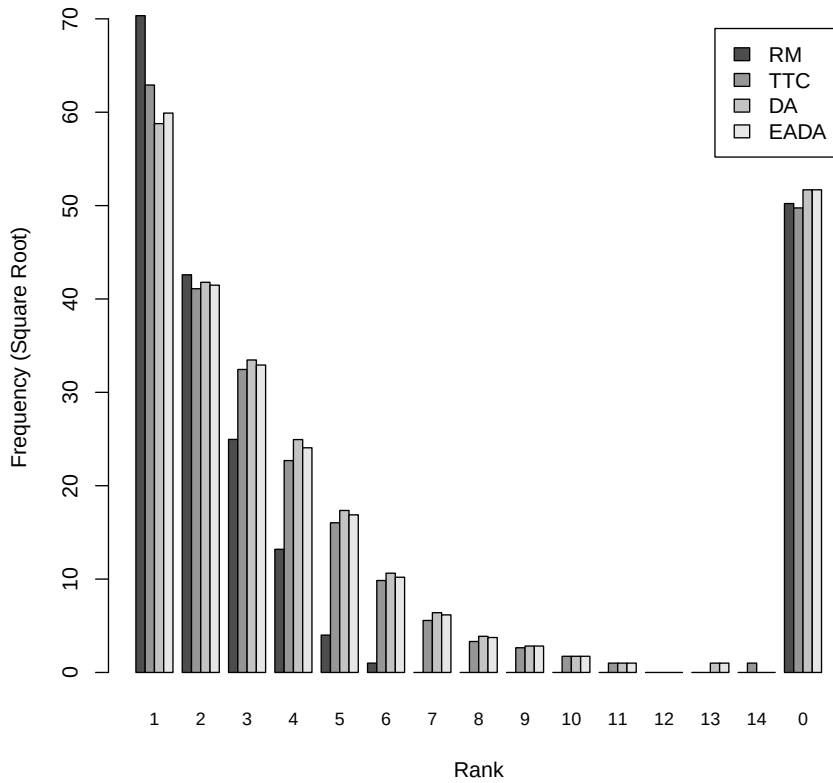


Figure 1: Rank distribution generated for 10,131 students in the secondary school admissions in Budapest using reported preferences. The last bar (0) denotes unassigned students.

we choose a problem at random in a market where all preferences and priorities are drawn uniformly at random. While the techniques extend to many-to-one markets, it is simpler to focus on markets with n students and n schools with just one seat. Old and seminal results show that Deferred Acceptance (a mechanism that is strategy-proof and stable) assigns the average student to a school ranked $\log n$, i.e. relatively close to the most preferred school with rank 1. But how much better could we do if we chose the best rank-minimizing mechanism, even if it is not strategy-proof? We showed that the best mechanism achieved a constant rank less than 2 for any n (even n equal to one billion!). Even better, the expected worst placement by the same algorithm was only $\log n$, quite smaller than that of Deferred Acceptance ($\log^2 n$). And despite the fact that this mechanism did not use priorities at all, it generates little envy (justified or not) because a lot of students (half, in fact) are assigned to their top choice. In fact, this rank minimizing mechanism generates less justified envy than Top Trading Cycles, even though TTC uses priorities and rank-minimizing does not. Testing these three mechanisms using Hungarian data showed that rank-minimizing could achieve significantly better allocations also in practice (see Figure 1).

This paper got me thinking about the large inefficiency of DA. While this negative aspect of DA was widely known, it was not known how pervasive it was or how frequently it emerged. In our Theoretical Economics paper, we answered both questions by studying the envy directed graph generated by DA. We show that as n grows, this envy digraph has a unique and giant strongly connected component, meaning that almost any two students were able to point at each other, thus participating in an improving cycle. This observation not only implied that the DA allocation was Pareto-inefficient with high probability, but also that almost all students could be simultaneously improved, because every set of cycles whose removal makes the graph acyclic covers almost all students. Our conclusion was robust to extending the model to many-to-one matching, to correlated preferences (in fact, the SCC

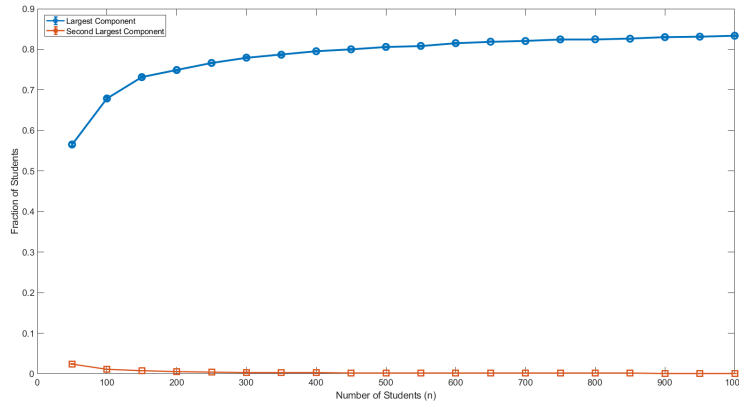


Figure 2: Average fraction of nodes in the largest (blue) and second-largest (red) SCCs (average over 2,000 random envy digraphs for each n).

becomes even larger then), and to moderate correlation in priorities. An illustration of the emergence of the unique giant SCC appears in Figure 2.

I find the large and likely DA inefficiency troublesome, so I have focused lately on mechanisms that produce allocations that Pareto-dominate that of DA. In our companion paper Ortega et al (2026, *JPE Micro*), we showed that none of these mechanisms could solve a big problem of DA in practice: in Chile and the UK it generates quite a lot of segregation. We showed that, unfortunately, every such mechanism preserves the tier of school to which a student is assigned, and furthermore, in very specific instances can generate quite a bad placement compared to mechanisms that do not dominate DA: almost every student may end up being worse off. On the positive side, in a recent unpublished paper we showed that any such mechanisms generates a significantly better placement in expectation, of average rank at most $\log \log n$ (Ortega et al., 2026a).

But could we implement any such mechanism? It would be more efficient, but necessarily it would require violating some priorities. Respecting priorities is a big reason why we like DA, so could we perhaps find a way to violate priorities in a justifiable manner? This is exactly what Pablo Arribillaga and I propose in a recent unpublished paper. Our idea is that an education authority can violate the priority of a student if he benefits or could never benefit, but not if he could benefit but didn't. Quite nicely, a justifiable improvement over DA always exists (if DA is inefficient), and one can find one such improvement in polynomial time that satisfies a constrained-efficiency property. Interestingly, our notion of controlled priority violation yields improvements that are unattainable with the alternative approach of controlled priority violations through ex-ante consent (Kesten, 2010). We showed that both approaches struggle to get full efficiency, and quantified their performance using publicly available school choice data from Chile's Sistema de Admisión Escolar.

To conclude, I also used the aforementioned random markets techniques to quantify the expected average rank of the Boston mechanism (which is of the same leading order as DA under truth-telling, and quite worse in equilibrium), see Ortega (2026a,b).

Merging Markets

My first paper (Ortega, 2018, *Journal of Mathematical Economics*) studied the problem of merging matching markets into a centralized clearinghouse. The obvious application was school district consolidation. I showed that there is a subtle trade-off for market participants, as integration yields more choice but also more competition. This paper has started a literature that has further explored which mechanisms encourage the integration of otherwise disjoint markets. (Ortega, 2019; Kumar et al., 2022) belong to this literature.

My first paper raised the question of whether integration of markets would actually yield welfare gains for most participants in practice, not only in theory. To verify this, in Klein, Aue, and Ortega (2024, *Games and Economic Behavior*) we analyze the district consolidation of 23 school districts in Budapest. We analyze just over 10,000 applications to secondary schools in two scenarios: applicants could apply either only to schools within their district or to any school in Budapest. We found very large gains from integration, equivalent to attending a school that was 5 kilometres closer to the students' addresses. The key methodological contribution was to estimate parents' preferences over schools and schools' priorities over students using only partial rank-ordered lists assuming both that the final allocation was stable and that students did not use weakly dominated strategies.

Great in Theory, Great in Practice?

The third strand of my research studies how well resource allocation mechanisms work in practice. My first paper on this studied cake-cutting: an interesting problem that gives insights on how to divide contested land: think of several agents who value different parts of a surface differently, and we can partition the surface arbitrarily in hope that we can produce an allocation that is either proportional or, even better, envy-free. It turns out doing this is simple when there are only two parties by a divide-and-choose method, but doing so for three agents requires a complex, sequential mechanism called Selfridge-Conway (doing so for four or more is realistically impossible, needing cutting the land over 200 times). This fancy mechanism is envy-free when agents report their valuations truthfully, but not necessarily when agents are strategic. Surprisingly, in a lab experiment (Kyropoulou, Ortega and Segal Halevi, 2022, *Games and Economic Behavior*) we found that even in cases with significant manipulations in practice the mechanism still worked amazingly well.

More recently I have studied the same question in the context of school choice mechanisms. The DA algorithm, as we have explained, is very inefficient, while the Boston mechanism is instead Pareto efficient under truthfulness, so tends to generate better placements. But interestingly, when considering equilibrium play, every equilibrium in the preference revelation game induced by Boston becomes weakly Pareto dominated by DA. This is what a large literature has found in experiments with undergrad students: DA is less manipulated than Boston. In a paper with Freer and Klein, recently accepted to the 27th Economics and Computation conference, we did a similar experiment but with parents versus students (this is the first school choice experiment actually using parents!). We found that parents manipulated at the same rate as students, but importantly, made significantly fewer obvious mistakes. We also tested two novel mechanisms: one called efficiency-adjusted DA, which performed quite poorly in all dimensions (see (Cerrone et al., 2024) for an experiment with the opposite findings in a much smaller market), and one called rank-minimizing that I studied in my GEB paper, which actually was much more efficient at the cost of generating many more blocking pairs.

More recently, in a brand new paper I study the comparison between the DA and Boston mechanisms once parents are allowed to appeal the placement received by the mechanism, as it happens in England and Northern Ireland, where over fifty thousand parents appeal each year, with one-in-five being successful. Theoretically, appeals allowed the Boston mechanism to support an equilibrium that Pareto dominates DA, and made the Boston mechanism less manipulable. This was quite exciting, and is exactly what we found in a pre-registered experiment (Cerrone et al., 2026): with appeals, Boston had fewer manipulations—even fewer than DA—and increased the efficiency gap between DA and Boston. We say increase the gap because surprisingly, we found that even without appeals, the Boston mechanism generated a better placement than DA. We really like this paper because it highlights that mechanisms aren't static, and must be evaluated as a whole because what happens after affects how parents behave in the first-stage of the mechanism.

Exciting Recent Work

1. *Theory+Data*. My paper on school choice with appeals that I just described is high on my list of most exciting working papers. It takes a real aftermarket that improves the placement of

thousands of students each year, and shows that it changes how parents report their preferences to the education authority.

2. *Theory*. My second most exciting project is related to my work on random markets but related to the allocation of *divisible* goods. The easiest way to solve this problem is through a random serial dictatorship: order agents randomly and let them choose their most preferred available object. While intuitive, this method has a serious flaw: it generates fractional assignments that aren't ordinaly efficient, meaning that agents want to swap shares over different objects. A beautiful eating algorithm called Probabilistic Serial solves this problem.

Yet, the literature has conflicting views of how relevant the ordinal efficiency problem is. On one hand, [Manea \(2009\)](#) showed that with n agents and n objects, the fraction of preference profiles for which the serial dictatorship is ordinaly efficient vanishes. On the other hand, [Che and Kojima \(2010\)](#) show that in replica markets, as the number of copies of each object becomes large, the probabilistic allocations generated by both mechanisms are asymptotically equivalent.

We help to reconcile these seemingly contradicting answers. In random iid markets, we show that PS's expected average rank is basically constant: it grows no faster than the iterated logarithm $\log^* n$. The iterated logarithm grows extraordinarily slowly: for example, $\log^* n = 4$ even when $n = 10^{80}$, the estimated number of atoms in the universe, showing that these two mechanisms diverge (serial dictatorship gives the much worse $\log n$), and that ordinal efficiency leads to significantly better expected allocations. When we allow for object replicas, we show that both mechanisms converge if and only if the number of object copies grows faster than the logarithm of object types. The working paper is now available ([Ortega and Zhao, 2026](#)).

3. *Data*. My final ongoing project is about school choice in Northern Ireland, which is quite specific. The country runs a fully centralized procedure, but schools have flexibility regarding how they prioritize students. Some schools prioritize students who rank them first (this has been illegal in England since 2007), some prioritize those who have ranked them first or second, and some give the same priority to all students no matter how they have ranked them. We track changes in admissions priorities of all 200 schools over 10 years, and show there are significant changes: schools frequently change their priority system. We find that when a school moves to the first-priority-first criterion, it receives 5% more first-choice applications, but roughly 5% fewer second-choice applications: i.e. changes in priority systems rarely change how many people rank the school, just the place where the school appears in their rankings.

Together, these projects reflect the direction of my research: combining economic theory, data and experiments to understand how allocation mechanisms perform in real institutions, and how they can be designed to work better.

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